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MAT 102

Introduction to Mathematical Proofs

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Function

A **function** f from a set A to a set B is a rule, that assigns to each element $a \in A$, a unique element $f(a) \in B$, called the **image** of a under f.

The notation $f: A \to B$ means that f is a function with **domain** A and **codomain** B.

The set $\{f(a): a \in A\}$, denoted as $f(A)$, is called the **range** (or the **image**) of the function f. Note that $f(A)$ is always a subset of the codomain B.

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Example:

We define a function $f: [-3,4] \to (-2,2)$ by $f(x) = \sin x$

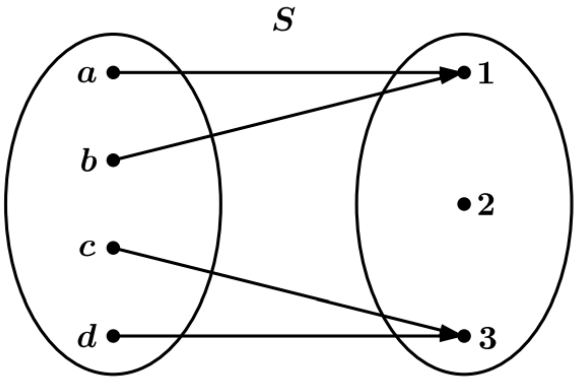
The image of $\frac{\pi}{2}$ is $f\left(\frac{\pi}{2}\right) = \sin\frac{\pi}{2} = 1$, and the image of $-\frac{\pi}{6}$ is $f\left(-\frac{\pi}{6}\right) = \sin\left(-\frac{\pi}{6}\right) = -1$

The codomain of f is $(-2,2)$

The range of f is $[-1,1]$

Example:

The diagram below defines a function $S: \{a, b, c, d\} \to \{1, 2, 3\}$.



The range of above function is $\{1, 3\}$, the codomain of above function is $\{1, 2, 3\}$

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Indicate whether each of the following defines a function:

(a) If S is the set of students in MAT102, define $f_1 : S \to \mathbb{N}$ which maps each person to their student number.

(b) Define $f_2 : \mathbb{Q} \to \mathbb{Z}$ by $f_2\left(\frac{a}{b}\right) = a$.

(c) Define $f_3 : \mathbb{N} \to \mathbb{N}$ by taking $n \in \mathbb{N}$ to the nearest prime to n.

(d) Let $\mathcal{B}$ be the set of all books, and define $f_4 : \mathcal{B} \to \mathbb{R}$ by taking $b \in \mathcal{B}$ to its word count.

(e) Define $f_5 : \mathbb{R} \to \mathbb{N}$ by $f_5(n) = n!$.

(f) Define $f_6 : \mathbb{N} \to \mathbb{R}$ by $f_6(n) = n!$.

(a) If S is the set of students in MAT102, define $f_1 : S \to \mathbb{N}$ which maps each person to their student number.
该映射将每个学生映射到其唯一的学号。因为每个学生都有一个独特的学号，所以对于集合S 中的每个元素，都对应唯一的输出值。因此，这确实定义了一个函数。

(b) Define $f_2 : \mathbb{Q} \to \mathbb{Z}$ by $f_2\left(\frac{a}{b}\right) = a$.
一个有理数可以有多种表示形式（例如，$\frac{1}{2}$ 也可以表示为$\frac{2}{4}$），导致相同的输入可能有多个不同的输出。因此，这不定义为一个函数。

(c) Define $f_3 : \mathbb{N} \to \mathbb{N}$ by taking $n \in \mathbb{N}$ to the nearest prime to n.
当距离相等的两个素数出现在n 两侧时。这种模糊性可能导致相同的输入n 对应多个输出，因此这不定义为一个函数。

(d) Let $\mathcal{B}$ be the set of all books, and define $f_4 : \mathcal{B} \to \mathbb{R}$ by taking $b \in \mathcal{B}$ to its word count.
每本书都有唯一的字数，因此对于每个输入（书）都对应唯一的输出（字数）。因此，这是一个函数。

(e) Define $f_5 : \mathbb{R} \to \mathbb{N}$ by $f_5(n) = n!$.
阶乘$n!$ 只在非负整数上有定义，而不适用于所有实数。因此，该映射在整个实数域$\mathbb{R}$ 上不成立，不能视为一个函数。

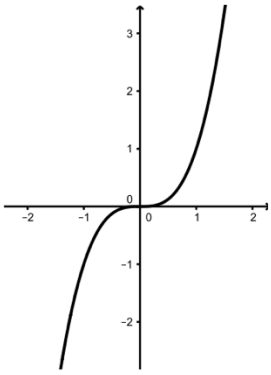
(f) Define $f_6 : \mathbb{N} \to \mathbb{R}$ by $f_6(n) = n!$. 对于所有自然数n，阶乘$n!$ 是定义明确的，并且每个输入值都对应唯一的输出值。因此，这确实定义了一个函数。

使用竖直线检测可以快速的判断一个图像是否是一个函数

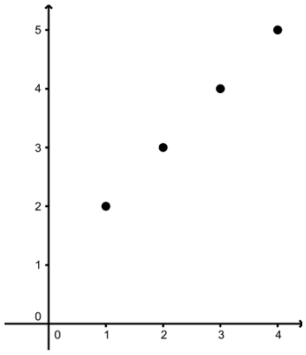
Graph of a function

The graph of a function $f: A \to B$ is the set $\{(a, f(a)): a \in A\}$.

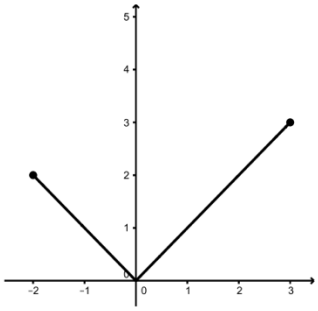
Example:



$f: \mathbb{R} \to \mathbb{R}$, $f(x) = x^3$



$p: \mathbb{N} \to \mathbb{N}$, $p(n) = n + 1$



$R: [-2, 3] \to \mathbb{R}$, $R(x) = |x|$

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Injective (one to one)

Codomain 中的每个值 y，都**至多**有一个 x 与之对应，使得$f(x) = y$ （水平线检测）

Surjective (on-to)

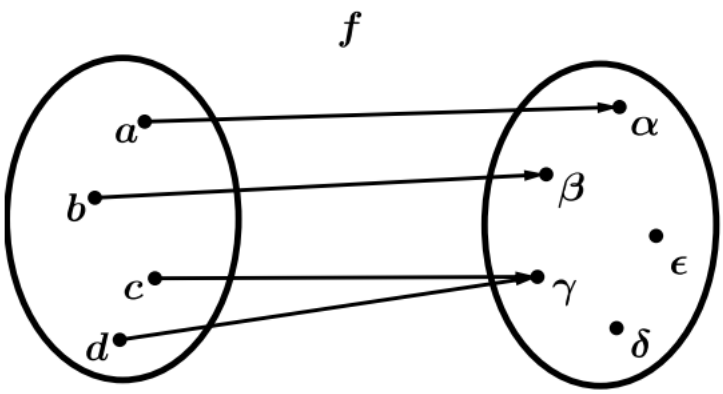
Codomain 中的每个值 y，都**至少**有一个 x 与之对应，使得$f(x) = y$

Bijective

一个 funciton 同时是 **Injective 和 Surjective，那么它是** Bijective

Question:

Is this function injective? Is this function surjective?



A B

Injective: B 中的每个值 y，都至多有一个 x，使得$f(x) = y$ （水平线检测）

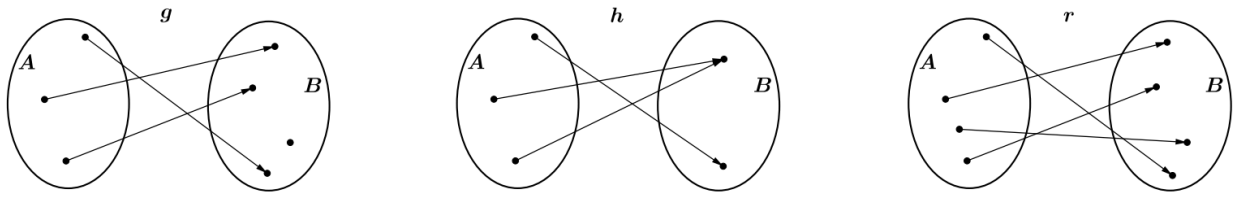
Surjective: B 中的每个值 y，都至少有一个 x，使得$f(x) = y$

In order to be able to invert a function, our initial function must be **both injective** and **surjective**.

Examples:

g, h and r are functions from a set A to another set B, given by the following diagrams

他们分别是 injective，surjective 还是 bijective?



The function g is **injective**

- as there are no two elements in A which are mapped to the same element in B
- g is not surjective as there is a "lonely" element in B, which is not the image of an element in A.

The function h is **surjective**,

- as every element in B is an image,
- but it is not one-to-one, since there are two arrows pointing to the same element in B.

The function r is **both an injection and a surjection**,

- as every element in B is the image of exactly one element in A.

Is following function injective and surjective

$$f: R \to R, f(x) = \frac{1}{1 + x^2}.$$

g is not an injection. 这个举反例就好，

存在一个$n_1 \neq n_2$，那么$g(n_1) = g(n_2)$

$$f(2) = f(-2) = \frac{1}{5}$$

g is not a surjection. 这个举反例就好，

因为$f(x) > 0$，因此没有能够让$f(x) = -1$ 的 x 值，因此它不是 **surjection.**

证明 injection

$$g: N \to N, g(n) = n^3 + 1.$$

g is an injection.

其实就是在证明，如果$n_1 \neq n_2$，那么$g(n_1) \neq g(n_2)$

或者证明：如果$g(n_1) = g(n_2)$，那么$n_1 = n_2$

We prove this by showing that distinct elements in the domain are mapped to distinct images. Indeed, if n_1, n_2 are two natural numbers, then

$$n_1^3 \neq n_2^3$$
$$n_1^3 + 1 \neq n_2^3 + 1$$
$$g(n_1) \neq g(n_2)$$

证明 **bijection**

$$h: R \setminus \{-1\} \to R \setminus \{1\}, \qquad h(x) = \frac{x}{x+1}.$$

h is a bijection. Here is the proof.

h is an injection.

To show that h is one-to-one, we start by assuming that $h(x_1) = h(x_2)$ for some $x1, x2 \neq -1$. We argue that $x_1 = x_2$ as follows.

$$h(x_1) = h(x_2)$$
$$\frac{x_1}{x_1+1} = \frac{x_2}{x_2+1}$$
$$x_1(x_2+1) = x_2(x_1+1)$$
$$x_1 = x_2$$

h is a surjective. 证明的思路是，对于任意的 y（在函数的值域中**），**总能找到一个 x 使得 $h(x) = y$

Now we show that h is surjective. Let $y \neq 1$. Our task is to show that $h(x) = y$ for some x in the domain of h.

We do that by solving the equation $h(x) = y$ for x:

$$\frac{x}{x+1} = y$$
$$x = (x+1)y$$
$$x = \frac{y}{1-y} \text{ (since } y \neq 1)$$

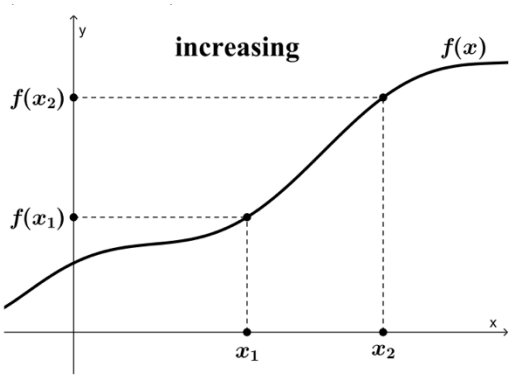
这样我们就说明了，你随便给我一个 y 值，我能告诉你，想要得到这个 y，x 是多少

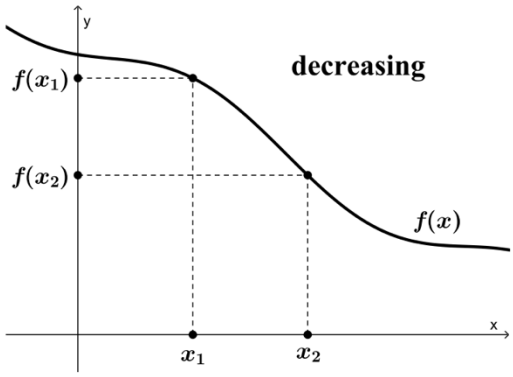
we showed that any y in the codomain of h has an 'x', which implies that h is surjective.

Monotone

We say that f is a strictly increasing (respectively decreasing) function, if for every $x_1 < x_2$ in A, we have $f(x_1) < f(x_2)$ (respectively $f(x_1) > f(x_2)$).

A function f is strictly monotone, if it is either strictly increasing or strictly decreasing.





Theorem

Let $A \subseteq R$. If $f : A \to R$ is a strictly monotone function, then f is injective.

(一定可以通过水平线测试)

Example:

Prove: the function $f : (-1, 1) \to R$, $f(x) = \frac{2x}{1-x^2}$ is injective

$$f'(x) = \frac{2(1+x^2)}{(1-x^2)^2} > 0 \qquad \text{for } -1 < x < 1.$$

Exercise:

Let $g: Z \times N \to Z \times N$ be given by $g(m,n) = (3-m, n^2+1)$

Part A: Is this function injective? Explain

Part B: Is this function surjective? Explain

(a) Let us show that the function is injective.

For this, we assume that $g(m_1, n_1) = g(m_2, n_2)$ for some $n_1, n_2 \in \mathbb{N}$, $m_1, m_2 \in \mathbb{Z}$. This implies that

$$(3 - m_1, n_1^2 + 1) = (3 - m_2, n_2^2 + 1),$$

$$n_1^2 + 1 = n_2^2 + 1 \text{ and } 3 - m_1 = 3 - m_2,$$

$$n_1^2 = n_2^2 \text{ and } m_1 = m_2,$$

$$|n_1| = |n_2| \text{ and } m_1 = m_2,$$

as both $n_1, n_2 \in \mathbb{N}$. Therefore, $(m_1, n_1) = (m_2, n_2)$. Hence, g is injective.

(b) Let us show that g is not surjective.

It is enough to show that, for example, $(0,1) \in \mathbb{Z} \times \mathbb{N}$ but not in $g(\mathbb{Z} \times \mathbb{N})$. By contradiction, if $(0,1) = g(m,n)$ for some $(m,n) \in \mathbb{Z} \times \mathbb{N}$. Then $(0,1) = (3-m, n^2+1)$. It follows that $1 = n^2 + 1$ and $0 = 3 - m$. The first equality implies that $n^2 = 0$, a contradiction.

Exercise:

Let $h: Z \times Z \to Z \times Z$ be given by $h(x, y) = (2xy, 7x - 3y)$

Part A: Is this function injective? Explain

Part B: Is this function surjective? Explain

(a) h is **not** an injective function since $h(-3, 0) = (0, -21) = h(0, 7)$ while $(-3, 0) \neq (0, 7)$.

(b) h is **not** a surjective function. It is sufficient to show that there exists an element in $\mathbb{Z} \times \mathbb{Z}$ which is not in $h(\mathbb{Z} \times \mathbb{Z})$. This would imply that $\mathbb{Z} \times \mathbb{Z}$ is **not** a subset of $h(\mathbb{Z} \times \mathbb{Z})$.

By contradiction. Assume that $(1, 0) = (2xy, 7x - 3y)$ where $x, y \in \mathbb{Z}$. It follows that $1 = 2xy$. However, $2xy$ is an even integer while 1 is odd. This is a contradiction.

(c) Since h is not injective (and not surjective), it cannot be a bijection.

1. (10 marks) Let $f : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ be the function given by $f(x) = x - \frac{1}{x}$.

(a) (4 marks) Is f injective? Provide a proof or a counterexample.

$f(x)$ is not injective.

For example, if $x_1 = 5$, $x_2 = -\frac{1}{5}$, then

$f(x_1) = 5 - \frac{1}{5}$ and $f(x_2) = -\frac{1}{5} + 5$.

Hence, $f(x_1) = f(x_2)$

(b) (6 marks) Is f surjective? Provide a proof or a counterexample.

$f(x)$ is surjective.

Choose any $y \in \mathbb{R}$. Set $y = x - \frac{1}{x}$. Solve for x.

$y = \frac{x^2 - 1}{x}$, $yx = x^2 - 1$, $x^2 - yx - 1 = 0$

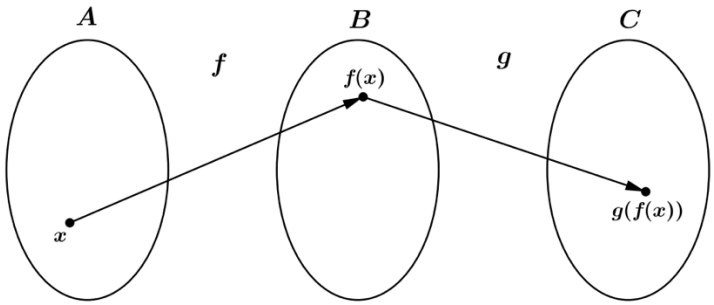
Since $D = (-y)^2 + 4 = y^2 + 4 > 0$, the equation has 2 distinct (nonzero) solutions.

This proves surjectivity.

Composition

The **composition** of g with f, denoted as $g \circ f$, is the function from A to C, given by

$$g \circ f(x) = g(f(x)) \qquad \text{for } x \in A.$$



Is the order of Composition matter? If not, give me a counter example

Composition 的顺序重要吗？如果不重要，请给我一个反例

Consider the functions $f, g\colon \mathbb{R} \to \mathbb{R}$, given by

$$f(x) = \frac{1}{1+x^2} \qquad \text{and} \qquad g(x) = e^x.$$

If we compose g with f, we obtain the function $g \circ f(x) = g(f(x)) = g\left(\frac{1}{1+x^2}\right) = e^{\frac{1}{1+x^2}}$.

We can also compose f with g (i.e., apply g first), to get $f \circ g(x) = f(g(x)) = f(e^x) = \frac{1}{1+(e^x)^2}$.

As we can see, $g \circ f \neq f \circ g$, which means that when we compose two functions, **the order of composing matters**.

假设$f: A \to B$ and $g: B \to C$ be two functions.

我能写$g \circ f$吗?

我能写$f \circ g$吗?

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Relation between composition

Example

Let $f: A \to B$ and $g: B \to C$ be two functions. Prove that if $g\ o\ f$ is injective, then so is f.

证明： 如果 $g\ o\ f$ 是 injective，那么f一定是 injective

如果 $f(x_1) = f(x_2)$, then, by applying the function g on both sides of this equality, we get

$$g(f(x_1)) = g(f(x_2)))$$
$$g\ o\ f(x_1) = g\ o\ f(x_2)$$

Since $g\ o\ f$ is injective, thus, from $g\ o\ f(x_1) = g\ o\ f(x_2)$, we have that $x_1 = x_2$

因此我们得出$f(x_1) = f(x_2) \to x_1 = x_2$

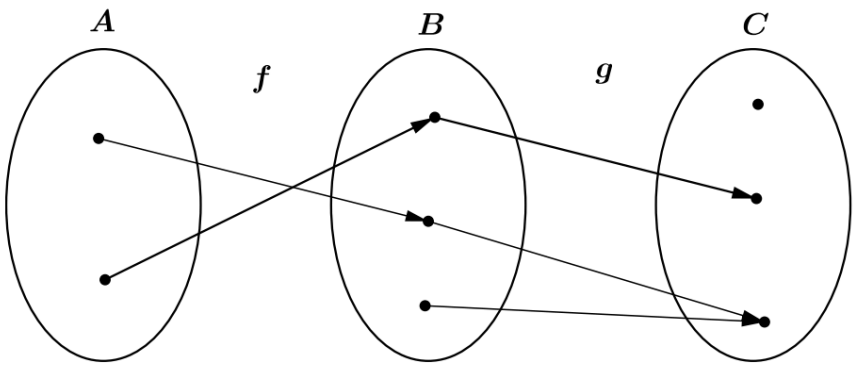
This completes the proof of injectivity of f

Example

If the composition $g \circ f$ is injective, must also g be injective?

Answer

不一定，下图就是一个例子



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定理

The composition of two injections is an injection

Let $f: A \to B$ and $g: B \to C$ be two injections. Our task is to prove that $g \circ f: A \to C$ is also a injection.

If $(g \circ f)(x_1) = (g \circ f)(x_2)$ then:

$$g(f(x_1)) = g(f(x_2))$$

Since g is injective, this implies:

$$f(x_1) = f(x_2)$$

Since $f(x_1) = f(x_2)$ and f is injective, we have:

$$x_1 = x_2$$

which shows that $g \circ f$ is injective.

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定理

The composition of two surjections is a surjection.

Proof

Let $f : A \to B$ and $g : B \to C$ be two surjections. Our task is to prove that $g \circ f : A \to C$ is also surjective

Let $z \in C$ be an arbitrary element.

As g is surjective, $z = g(y)$ for some $y \in B$. Similarly, f is surjective, and so $y = f(x)$ for some $x \in A$.

Over all, we have

$$z = g(y) = g(f(x)) = g \circ f(x)$$

which shows that $g \circ f$ is surjective.

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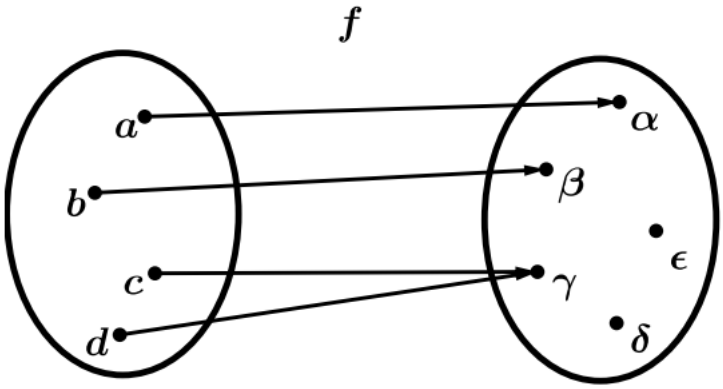
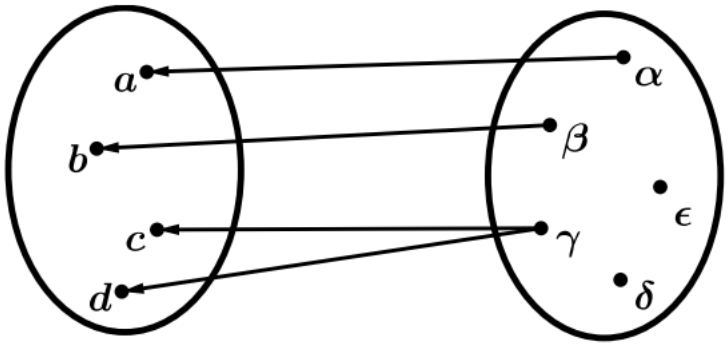
Inverse

Consider a function f : {a, b, c, d} → {α, β, γ, δ, ϵ}, given by the following diagram



Does this function have inverse?



In the new diagram, both c and d are assigned to the element γ.

Functions **cannot assign two images** to an element in the domain.

In the new diagram, no elements are assigned to δ and ϵ. However, a function **must assign an image** to every element in its domain.

Left-inverse and right-inverse

Let $f: S \to T$ be a function.

$g: T \to S$ is a **left-inverse** of f if $g(f(s)) = s$ for all $s \in S$,

$g: T \to S$ is a **right-inverse** of f if $f(g(t)) = t$ for all $t \in T$,

Example:

Does $f(x) = \sqrt{x}$ has **left-inverse or right-inverse**

If we let $g(x) = x^2$

$$g(f(s)) = (\sqrt{x})^2 = x$$

$$f(g(s)) = \sqrt{(x)^2} = |x|$$

因此我们说它是 **left-inverse** 而不是 **right-inverse**

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Example:

Does $f(x) = x^2$ has **left-inverse or right-inverse**

Note:

快速判断的方法:

surjective -> right-inverse

injectvie -> left-inverse

定理

If $f: A \to B$ and $g: B \to C$ are two bijections, then $(g\ o\ f)^{-1} = f^{-1}\ o\ g^{-1}$.

Proof

Our task is to show that they are equal to each other. I.e., we need to prove that $(g\ o\ f)^{-1}(z) = f^{-1}\ o\ g^{-1}(z)$ for any $z \in C$

假设$f(x) = y, g(y) = z$, and denote $y = g^{-1}(z)$ and $x = f^{-1}(y)$.

Therefore,

$$f^{-1}\ o\ g^{-1}(z) = f^{-1}\left(g^{-1}(z)\right) = f^{-1}(y) = x$$

On the other hand

$$g\ o\ f(x) = g\big(f(x)\big) = g(y) = z$$

Hence:

$$(g\ o\ f)^{-1}(z) = x$$

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定理

The composition of two bijections is a bijection.

Exercise:

Let $f : A \to B$ be an arbitrary function.

Prove that if f is a bijection (and hence invertible), then $f^{-1}(f(x)) = x$ for all $x \in A$, and $f(f^{-1}(x)) = x$ for all $x \in B$.

由函数值的定义，令 $y = f(x) \in B$。
因为 f 是双射，故对每个 $y \in B$ 存在且仅存在一个 $a \in A$ 使得 $f(a) = y$。逆函数 f^{-1} 就是把这个 y 送回到该唯一的 a。

现在对我们的 $y = f(x)$ 应用逆函数定义：

$$f^{-1}(f(x)) = f^{-1}(y) = a,$$

其中 a 是满足 $f(a) = y = f(x)$ 的唯一元素。由"唯一性"，这个 a 必须就是 x。故

$$f^{-1}(f(x)) = x.$$

取任意 $x \in B$。因为 f 是**满射**，存在 $a \in A$ 使得 $f(a) = x$。由逆函数的定义，$f^{-1}(x)$ 就是这个把 x 送回的唯一元素，也即 $f^{-1}(x) = a$。于是

$$f\big(f^{-1}(x)\big) = f(a) = x.$$

两式均证毕，故

$$f^{-1} \circ f = \mathrm{id}_A, \qquad f \circ f^{-1} = \mathrm{id}_B\,.$$

Exercise:

Let $f: A \to R$ be a function (where A is a set). Define a new function $g: A \to R$ by $g(x) = 3 \cdot [f(x)]^2 + 1$. Prove that if g is injective, then f is injective.

设 g 单射。任取 $x_1, x_2 \in A$ 且满足

$$f(x_1) = f(x_2).$$

则

$$g(x_1) = 3[f(x_1)]^2 + 1 = 3[f(x_2)]^2 + 1 = g(x_2).$$

由 g 的单射性，得 $x_1 = x_2$。因此由定义，f 是单射。□

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